Bayblend® T88 GF-20

Standard grades / Glass fiber reinforced	Rubber modified (PC+SAN) blend; 20% glass fiber filled; Vicat/B 120 temperature = 130 °C; optimized heat ageing- and UV-stability; very good flow; tensile modulus = 7200 MPa; good heat resistance
ISO Shortname	PC+SAN-I-GF20

	Property	Test Condition	Unit	Standard	typical Value
Rheological properties					
C	Melt volume-flow rate	260 °C/ 5 kg	cm³/10 min	ISO 1133	14
	Melt viscosity	1000 s^{-1}/ 260 °C	Pa·s	b.o. ISO 11443-A	205
	Molding shrinkage, parallel	150x105x3 mm/ 260 °C / MT 80 °C	%	b.o. ISO 2577	0.2 - 0.4
	Molding shrinkage, normal	150x105x3 mm/ 260 °C / MT 80 °C	%	b.o. ISO 2577	0.3 - 0.5
Mechanical properties (23 °C/50 % r. h.)					
C	Tensile modulus	1 mm/min	MPa	ISO 527-1,-2	7200
	Yield stress	5 mm/min	MPa	ISO 527-1,-2	120
	Yield strain	5 mm/min	%	ISO 527-1,-2	2.4
C	Stress at break	5 mm/min	MPa	ISO 527-1,-2	120
C	Strain at break	5 mm/min	%	ISO 527-1,-2	2.4
	Izod impact strength	23 °C	kJ/m²	ISO 180/U	38
	Izod impact strength	-30 °C	kJ/m²	ISO 180/U	38
	Izod notched impact strength	23 °C	kJ/m²	ISO 180/A	8.0
	Izod notched impact strength	-30 °C	kJ/m²	ISO 180/A	8.0
Thermal properties					
C	Temperature of deflection under load	1.80 MPa	°C	ISO 75-1,-2	119
C	Temperature of deflection under load	0.45 MPa	°C	ISO 75-1,-2	129
C	Vicat softening temperature	50 N; 50 °C/h	°C	ISO 306	128
	Vicat softening temperature	50 N; 120 °C/h	°C	ISO 306	130
C	Coefficient of linear thermal expansion, parallel	23 to 55 °C	10^{-4}/K	ISO 11359-1,-2	0.3
C	Coefficient of linear thermal expansion, normal	23 to 55 °C	10^{-4}/K	ISO 11359-1,-2	0.65
C	Burning behavior UL 94 [UL recognition]	0.85 mm	Class	UL 94	HB
Electrical properties (23 °C/50 % r. h.)					
C	Relative permittivity	100 Hz	-	IEC 60250	3.3
C	Relative permittivity	1 MHz	-	IEC 60250	3.2
C	Dissipation factor	100 Hz	10^{-4}	IEC 60250	25
C	Dissipation factor	1 MHz	10^{-4}	IEC 60250	85
C	Volume resistivity		Ohm·m	IEC 60093	1E14
C	Surface resistivity		Ohm	IEC 60093	1E17
C	Electrical strength	1 mm	kV/mm	IEC 60243-1	35
C	Comparative tracking index CTI	Solution A	Rating	IEC 60112	150
Other properties (23 °C)					
C	Water absorption (saturation value)	Water at 23 °C	%	ISO 62	0.4
C	Water absorption (equilibrium value)	23 °C; 50 % r. h.	%	ISO 62	0.2
C	Density		kg/m³	ISO 1183-1	1290
	Glass fiber content	Method A	%	b.o. ISO 3451-1	20
Processing conditions for test specimens					
C	Injection molding - Melt temperature		°C	ISO 294	260
C	Injection molding - Mold temperature		°C	ISO 294	80
C	Injection molding - Injection velocity		mm/s	ISO 294	540

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	Property	Test Condition	Unit	Standard	typical Value
Recommended processing and drying conditions					
	Melt temperatures		°C	-	260 - 280
	Standard Melt temperature		°C	-	270
	Barrel Temperatures - Rear		°C	-	230 - 240
	Barrel Temperatures - Middle		°C	-	235 - 245
	Barrel Temperatures - Front		°C	-	240 - 270
	Barrel Temperatures - Nozzle		°C	-	265 - 275
	Mold Temperatures		°C	-	70 - 90
	Hold Pressure (% of injection pressure)		%	-	50 - 75
	Plastic Back Pressure (specific)		bar	-	50 - 150
	Peripheral Screw Speed		m/s	-	0.05 - 0.2
	Shot-to-Cylinder Size		%	-	30 - 70
	Dry Air Drying Temperature		°C	-	95 - 110
	Dry Air Drying Time		h	-	4
	Moisture Content max. (%)		%	-	<= 0,02
	Vent Depth		mm	-	0.025 - 0.075

C These property characteristics are taken from the CAMPUS plastics data bank and are based on the international catalogue of basic data for plastics according to ISO 10350.

Impact properties: N = non-break, P = partial break, C = complete break

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Disclaimer

Information Impact properties

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Typical value

These values are typical values only. Unless explicitly agreed in written form, the do not constitute a binding material specification or warranted values. Values may be affected by the design of the mold/die, the processing conditions and coloring/pigmentation of the product. Unless specified to the contrary, the property values given have been established on standardized test specimens at room temperature.

General

The manner in which you use and the purpose to which you put and utilize our products, technical assistance and information (whether verbal, written or by way of production evaluations), including any suggested formulations and recommendations are beyond our control. Therefore, it is imperative that you test our products, technical assistance, information and recommendations to determine to your own satisfaction whether our products, technical assistance and information are suitable for your intended uses and applications. This application-specific analysis must at least include testing to determine suitability from a technical as well as health, safety, and environmental standpoint. Such testing has not necessarily been done by Covestro. Unless we otherwise agree in writing, all products are sold strictly pursuant to the terms of our standard conditions of sale which are available upon request. All information and technical assistance is given without warranty or guarantee and is subject to change without notice. It is expressly understood and agreed that you assume and hereby expressly release us from all liability, in tort, contract or otherwise, incurred in connection with the use of our products, technical assistance, and information. Any statement or recommendation not contained herein is unauthorized and shall not bind us. Nothing herein shall be construed as a recommendation to use any product in conflict with any claim of any patent relative to any material or its use. No license is implied or in fact granted under the claims of any patent. With respect to health, safety and environment precautions, the relevant Material Safety Data Sheets (MSDS) and product labels must be observed prior to working with our products.

Non Medical and non Food Contact Grade

This product is not designated for the manufacture of a pharmaceutical/medicinal product, medical device or of intermediate products for medical devices1). This product is also not registered for Covestro for the use in other specifically regulated applications, in particular applications requiring regulatory registration, approval or notification (e.g. including cosmetics, plant protection, food processing, food contact and others). If the intended use of the product is for the manufacture of a pharmaceutical, medical device or of intermediate products for medical devices or for other specifically regulated applications which may lead to a regulatory obligation of Covestro, Covestro must be contacted in advance to provide its agreement to sell such product for such purpose. Nonetheless, any determination as to whether a product is appropriate for use in a pharmaceutical, medical device or intermediate products for medical devices or for the use in other specifically regulated applications, must be made solely by the purchaser of the product without relying upon any representations by Covestro, irrespective of the existence of any regulatory obligation for the registration, approval or notification. 1) Please see the "Guidance on Use of Covestro Products in a Medical Application" document.

Recommended Processing and Drying Conditions

Barrel temperatures are valid for a standard 3-zone barrel. Temperature set-up for different barrel types may change according to configuration. Values for hold pressure as percentage of injection pressure may vary depending on, amongst others, part geometry, injection molding machine and injection mold. Drying conditions are for dry air dryers only. Drying times and drying temperatures may differ depending on valid dryer type. Further information is provided by your local Covestro support as well as in the following brochures: Injection Molding of High Quality Molded Parts - Drying; Determining the Dryness of Makrolon by TVI Test; The fundamentals of shrinkage in thermoplastics; Shrinkage and deformation of glass fiber reinforced thermoplastics [...]. https://www.plastics.covestro.com/Library/Overview.aspx

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